

AuraSense

Smart Space Monitor

"See the air you breathe."

User Manual · v1.0

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1. Overview

AuraSense is an indoor air quality monitor built on the M5Stack Core2 device. It measures temperature, humidity, air pressure, CO₂, TVOC, and motion — and sends the data to a cloud dashboard in real time.

The device works in two modes:

- Offline Mode — full sensor monitoring, LED alerts, and screen display without WiFi
- Online Mode — cloud upload, AI voice assistant, weather forecast, and voice alerts

2. Hardware & Sensors

2.1 Device

M5Stack Core2 for AWS Edukit — with a 320×240 touch screen, built-in microphone and speaker, and a 10-LED RGB strip.

2.2 Sensors

Port	Sensor	Measures
Port A (I2C)	SGP30	eCO ₂ (ppm), TVOC (ppb)
Port B (GPIO)	AS312 PIR	Motion detection
Port C (I2C)	ENV III (SHT30 + QMP6988)	Temperature, Humidity, Pressure

3. Getting Started

3.1 Powering On

Connect the device via USB-C. The screen will show the AuraSense logo and "Initializing..." while the sensors start up.

3.2 Connecting to WiFi

On first boot (or after a reset), the device shows a QR code. Scan it with your phone to open the WiFi setup page:

- Enter your WiFi network name and password
- Tap Submit — the device will show a success or error message
- On success, the device restarts automatically and connects

 *To change WiFi later: go to the Settings page (long press Button A) and scan the QR code again.*

3.3 What Happens After Connecting

- Time syncs automatically from the internet (NTP)
- 5-day weather forecast loads immediately
- Sensor data starts uploading to the cloud every 30 seconds

4. Navigation

4.1 Buttons

Button	Short Press	Long Press (600ms+)
Button A (Left)	Previous page	Open Settings
Button B (Center)	Go to Home page	—
Button C (Right)	Next page	Voice Assistant

4.2 Pages

- Home — current temperature, humidity, outdoor weather, and AI advice
- Sensors — all sensor readings in detail
- Weather — 5-day forecast with temperature, humidity, wind, and rain chance
- Settings — WiFi setup

5. Screen Brightness

The screen adjusts automatically based on activity:

State	Brightness	When
Active	100%	Motion detected or user interaction
Dimmed	20%	After 60 seconds with no movement
Off	0%	After 120 seconds with no movement

To wake the screen: move in front of the device, touch the screen, or press any button.

6. LED Alert System

The 10 RGB LEDs on top of the device show air quality at a glance — even with the screen off.

LED Color	Flash Speed	eCO ₂ / TVOC	Humidity
Off	—	Normal	30% – 65%
Yellow	500ms	CO ₂ > 800 ppm TVOC > 220 ppb	< 30% or > 65%
Red	250ms	CO ₂ > 1500 ppm TVOC > 660 ppb	< 20% or > 75%

 *The LED alert works even when the screen is off. If the LEDs flash red at night, open a window immediately.*

7. Automatic Voice Alerts

When air quality reaches the danger level (red LED), AuraSense will:

- Wait 10 seconds to confirm the reading is stable
- Play a spoken alert through the built-in speaker
- Repeat the alert every 30 seconds until the air quality improves

 *Voice alerts fire even when the screen is off — so you are protected while sleeping.*

8. Hourly Announcement

Every hour, AuraSense plays a short spoken summary of the current indoor air quality and outdoor weather conditions.

The announcement is skipped if the screen has been off for a while, assuming you are away.

9. Voice Assistant

You can ask AuraSense questions directly using your voice.

9.1 How to Use

- Hold Button C (right button) — the screen shows "REC"
- Ask your question
- Release the button — the device uploads your recording
- Wait a few seconds — the answer plays through the speaker

9.2 Example Questions

- "What is the current air quality?"
- "Should I open a window?"
- "What is the weather like in Lausanne tomorrow?"
- "Is the CO2 level safe right now?"
- "What was the temperature at home last night?"

 *The assistant remembers the conversation context, so you can ask follow-up questions naturally.*

10. Web Dashboard

Access the live dashboard from any browser at the following address:

<http://34.62.132.20:8501>

10.1 Features

- Real-time sensor data with 24-hour trend charts
- Outdoor weather — temperature, humidity, wind, and forecast
- Anomaly log — all past warnings and danger events with timestamps
- Trend analysis — temperature, humidity, CO2, and TVOC over 1h, 24h, or 7 days
- 7-day daily summary and hourly motion heatmap
- AI recommendations — updated every 10 minutes

10.2 Online Status

The dashboard shows a green "Live" indicator when the device is connected. If the device has not uploaded data for more than 2 minutes, it shows "Offline" and the last known values.

11. Troubleshooting

Problem	Solution
Screen stays off	Wave your hand in front of the device, or press any button
WiFi not connecting	Check your password. Go to Settings (long press A) and scan the QR code again
"No reply" after voice	The server may be busy. Wait a moment and try again
Dashboard shows Offline	Check that the device is powered on and WiFi is connected
LED keeps flashing red	Ventilate the room — open windows or doors
Time is wrong	The device syncs time automatically when WiFi connects. Check your WiFi connection

12. Technical Specifications

Item	Details
Device	M5Stack Core2 for AWS Edukit (ESP32)
Firmware	UIFlow2.0 V2.4.4 / MicroPython
Sensor update rate	Every 3 seconds
Cloud upload interval	Every 30 seconds (every 5s during alerts)
Screen dim	After 60 seconds idle
Screen off	After 120 seconds idle
AI model	Google Gemini
TTS	Edge TTS (en-US-JennyNeural)
Cloud storage	Google BigQuery
Dashboard	Streamlit on Google Cloud

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github.com/embarky/AuraSense-M5StackCore2